

Engineering Physics — Module 1

Important Numerical Questions & Step-by-Step Solutions • Quantum Mechanics

Question 1

de Broglie Wavelength & Momentum

Calculate the momentum of an electron and the de Broglie wavelength associated with it if its kinetic energy is **1.5 keV**.

SOLUTION

Given Data:

Kinetic Energy, $E_k = 1.5 \text{ keV} = 1.5 \times 10^3 \text{ eV} = 1.5 \times 10^3 \times 1.602 \times 10^{-19} \text{ J} = 2.403 \times 10^{-16} \text{ J}$

Mass of electron, $m = 9.109 \times 10^{-31} \text{ kg}$

Planck's constant, $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$

Step 1: Calculate Linear Momentum (p)

The relation between kinetic energy and momentum is $E_k = p^2 / (2m) \Rightarrow p = \sqrt{2 m E_k}$

$$p = \sqrt{2 \times 9.109 \times 10^{-31} \text{ kg} \times 2.403 \times 10^{-16} \text{ J}}$$

$$p = \sqrt{4.3778 \times 10^{-46}} = 2.092 \times 10^{-23} \text{ kg}\cdot\text{m/s}$$

Step 2: Calculate de Broglie Wavelength (λ)

Using de Broglie equation: $\lambda = h / p$

$$\lambda = (6.626 \times 10^{-34} \text{ J}\cdot\text{s}) / (2.092 \times 10^{-23} \text{ kg}\cdot\text{m/s})$$

$$\lambda = 3.167 \times 10^{-11} \text{ m} = 0.03167 \text{ nm} = 0.3167 \text{ \AA}$$

Final Answer: Momentum = $2.092 \times 10^{-23} \text{ kg}\cdot\text{m/s}$ | de Broglie Wavelength = $0.3167 \text{ \AA}$ (0.0317 nm)

Question 2

Hydrogen Atom Ground State

The velocity of an electron in the ground state of a hydrogen atom is **$2.19 \times 10^6 \text{ m/s}$** . Calculate the wavelength of de Broglie waves associated with its motion.

SOLUTION

Given Data:

Velocity of electron, $v = 2.19 \times 10^6 \text{ m/s}$

Mass of electron, $m = 9.109 \times 10^{-31} \text{ kg}$

Planck's constant, $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$

Step 1: Apply de Broglie Relation

$$\lambda = h / (m v)$$

Step 2: Substitute Values & Calculate

$$\lambda = (6.626 \times 10^{-34}) / (9.109 \times 10^{-31} \times 2.19 \times 10^6)$$

$$\lambda = (6.626 \times 10^{-34}) / (1.9949 \times 10^{-24})$$

$$\lambda = 3.321 \times 10^{-10} \text{ m} = 0.3321 \text{ nm} = 3.321 \text{ \AA}$$

Final Answer: de Broglie Wavelength $\lambda = 3.321 \text{ \AA}$ ($3.321 \times 10^{-10} \text{ m}$)

Question 3

Accelerated Charged Particle

Evaluate the de Broglie wavelength of a Helium nucleus (α -particle) accelerated through a potential difference of 500 V.

SOLUTION

Given Data:

Accelerating potential, $V = 500$ V

Charge of helium nucleus (He^{2+}), $q = 2e = 2 \times 1.602 \times 10^{-19}$ C = 3.204×10^{-19} C

Mass of helium nucleus, $m \approx 4 \times 1.6726 \times 10^{-27}$ kg = 6.643×10^{-27} kg

Planck's constant, $h = 6.626 \times 10^{-34}$ J·s

Step 1: Kinetic Energy Acquired by α -particle

$$E_k = q V = (3.204 \times 10^{-19} \text{ C}) \times 500 \text{ V} = 1.602 \times 10^{-16} \text{ J}$$

Step 2: Calculate de Broglie Wavelength

$$\lambda = h / \sqrt{2 m q V}$$

$$\text{Denominator} = \sqrt{(2 \times 6.643 \times 10^{-27} \times 1.602 \times 10^{-16})} = \sqrt{(2.1284 \times 10^{-42})} = 1.4589 \times 10^{-21} \text{ kg}\cdot\text{m/s}$$

$$\lambda = (6.626 \times 10^{-34}) / (1.4589 \times 10^{-21})$$

$$\lambda = 4.541 \times 10^{-13} \text{ m} = 0.000454 \text{ nm} = 0.00454 \text{ \AA}$$

Final Answer: Wavelength $\lambda = 4.541 \times 10^{-13}$ m (0.00454 $\text{\AA}$)

Question 4

Photon vs. Particle Energy Comparison

Compare the energy of a photon with the kinetic energy of a neutron when both are associated with a wavelength of 1 $\text{\AA}$ (1.0×10^{-10} m). Given mass of neutron = 1.675×10^{-27} kg.

SOLUTION

Given Data:

Wavelength, $\lambda = 1 \text{ \AA} = 1.0 \times 10^{-10}$ m

Mass of neutron, $m_n = 1.675 \times 10^{-27}$ kg

Speed of light, $c = 3.0 \times 10^8$ m/s, Planck's constant, $h = 6.626 \times 10^{-34}$ J·s

Step 1: Energy of Photon (E_γ)

Photons travel at speed c : $E_\gamma = h c / \lambda$

$$E_\gamma = (6.626 \times 10^{-34} \times 3.0 \times 10^8) / (1.0 \times 10^{-10}) = 1.9878 \times 10^{-15} \text{ J}$$

$$\text{In eV: } E_\gamma = (1.9878 \times 10^{-15}) / (1.602 \times 10^{-19}) = 12,408 \text{ eV (12.41 keV)}$$

Step 2: Kinetic Energy of Neutron (E_n)

For a non-relativistic neutron: $E_n = p^2 / (2 m_n) = h^2 / (2 m_n \lambda^2)$

$$E_n = (6.626 \times 10^{-34})^2 / [2 \times 1.675 \times 10^{-27} \times (10^{-10})^2]$$

$$E_n = (4.3904 \times 10^{-67}) / (3.35 \times 10^{-47}) = 1.3106 \times 10^{-20} \text{ J}$$

$$\text{In eV: } E_n = (1.3106 \times 10^{-20}) / (1.602 \times 10^{-19}) = 0.0818 \text{ eV (81.8 meV)}$$

Step 3: Calculate Energy Ratio (E_γ / E_n)

$$\text{Ratio} = (1.9878 \times 10^{-15} \text{ J}) / (1.3106 \times 10^{-20} \text{ J}) \approx 151,675$$

Final Answer: The energy of the photon is $\sim 1.517 \times 10^5$ times (151,675 $\times$) greater than that of the neutron.

Question 5

KE, Phase & Group Velocity

An electron has a de Broglie wavelength of **2.0 nm**. Find its kinetic energy, phase velocity (v_p), and group velocity (v_g). Given rest mass energy of electron = **511 keV**.

SOLUTION

Given Data:

de Broglie wavelength, $\lambda = 2.0 \text{ nm} = 2.0 \times 10^{-9} \text{ m}$

Rest mass energy, $E_0 = m_0 c^2 = 511 \text{ keV} = 511 \times 10^3 \times 1.602 \times 10^{-19} \text{ J} = 8.1862 \times 10^{-14} \text{ J}$

Rest mass of electron, $m_0 = 9.109 \times 10^{-31} \text{ kg}$

Step 1: Calculate Linear Momentum & Group Velocity (v_g)

Momentum $p = h / \lambda = (6.626 \times 10^{-34}) / (2.0 \times 10^{-9}) = 3.313 \times 10^{-25} \text{ kg}\cdot\text{m/s}$

Group velocity equals particle velocity ($v_g = v$):

$$v_g = p / m_0 = (3.313 \times 10^{-25}) / (9.109 \times 10^{-31}) = 3.637 \times 10^5 \text{ m/s}$$

Step 2: Calculate Kinetic Energy (E_k)

Since $v_g \ll c$, non-relativistic formula applies:

$$E_k = p^2 / (2 m_0) = (3.313 \times 10^{-25})^2 / (2 \times 9.109 \times 10^{-31}) = 6.025 \times 10^{-20} \text{ J}$$

$$\text{In eV: } E_k = (6.025 \times 10^{-20}) / (1.602 \times 10^{-19}) = 0.376 \text{ eV}$$

Step 3: Calculate Phase Velocity (v_p)

Relativistic Total Energy formulation: $v_p = c^2 / v_g$

$$v_p = (3.0 \times 10^8)^2 / (3.637 \times 10^5) = 2.474 \times 10^{11} \text{ m/s}$$

(Note: In pure non-relativistic wave mechanics where $E = E_k$, $v_p = \frac{1}{2} v_g = 1.818 \times 10^5 \text{ m/s}$).

Final Answer: KE = 0.376 eV ($6.025 \times 10^{-20} \text{ J}$) | Group Velocity $v_g = 3.637 \times 10^5 \text{ m/s}$ | Phase Velocity $v_p = 2.474 \times 10^{11} \text{ m/s}$

Quick Formula Reference for Module 1 Numerical Exams

$$\bullet \lambda = h / p = h / \sqrt{2mE}$$

$$\bullet v_g = d\omega / dk = v$$

$$\bullet \lambda_e = 12.26 / \sqrt{V} \text{ \AA}$$

$$\bullet v_p = \omega / k = E / p = c^2 / v_g$$